SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 24

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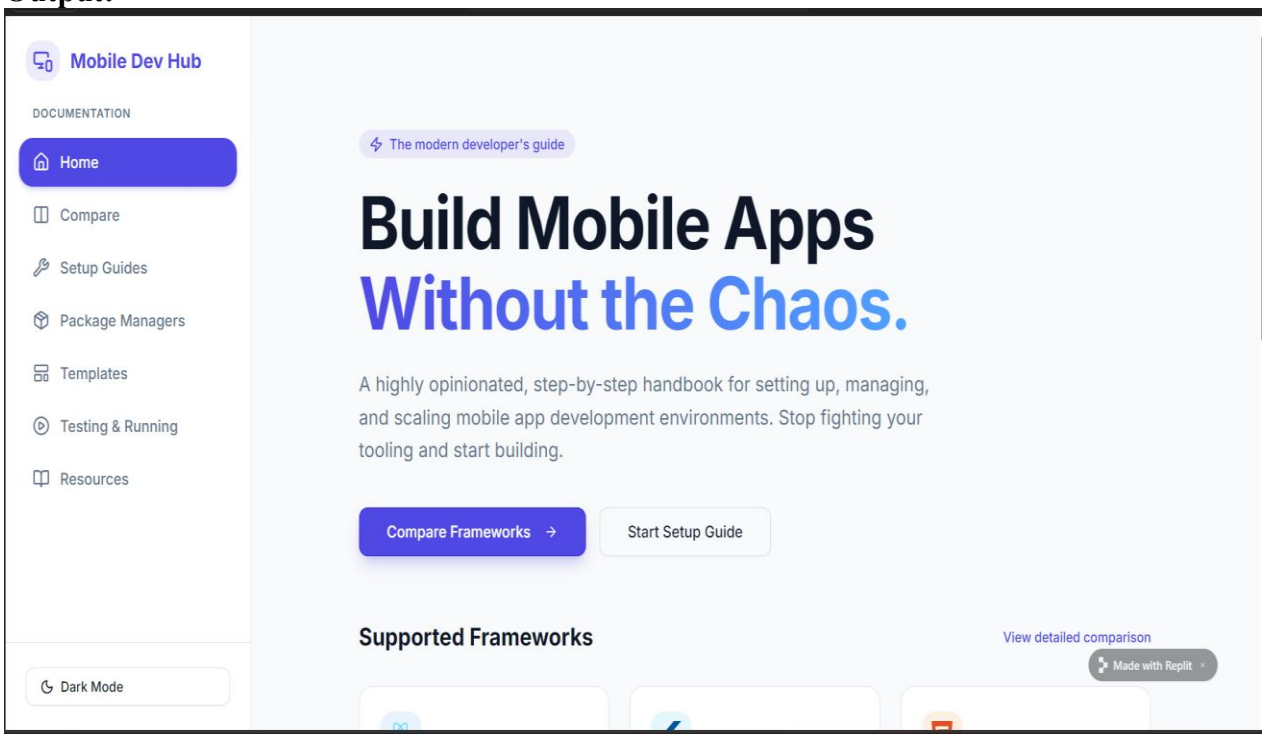
Batch No. : 22

Task 1 – Environment Setup
• Create a free Replit account.
• Explore the mobile app development templates (React Native / Flutter / HTML5 hybrid).
• Install required packages or dependencies in the Replit environment.

Prompt:-
To set up a mobile app development environment, first create a free account on Replit by signing up with your email or Google account. After logging in, explore the available templates for mobile app development such as React Native, Flutter, and HTML5 hybrid applications. Choose a suitable template and create a new project workspace. Then, install the required packages and dependencies using the built-in package manager or shell in Replit. Finally, run and test the application directly within the Replit environment to ensure it works properly.

Output:-



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Mobile Dev Hub

DOCUMENTATION

- Home
- Compare
- Setup Guides
- Package Managers
- Templates
- Testing & Running
- Resources

Dark Mode

Choosing the right framework is the most critical decision for your mobile project. Here's an honest comparison of the big three.

	React Native	Flutter	HTML5 Hybrid
	Learn once, write anywhere.	Build native interfaces in record time.	Web tech wrapped in a native shell.
Language	JavaScript / TypeScript	Dart	HTML / CSS / JS
UI Components	Native UI Components	Custom rendering engine	Web DOM (DOM elements)
Performance	Near Native	Excellent	Good (Webview)
Learning Curve	Low (if you know React)	Medium (requires Dart)	Very Low
Hot Reload	✓	✓	✓
Ecosystem	Massive (NPM)	Growing rapidly (pub.dev)	Web Ecosystem
App Size	Medium	Large	Small to Medium
Pros	• Huge community • Code sharing with web • Over-the-air updates	• Consistent UI across platforms • Excellent performance • Great documentation	• Leverage existing web team • Extremely fast development • Access to all web libraries
Cons	• Bridge performance bottleneck • Complex native module linking	• Must learn Dart • Larger app sizes • Fewer third-party libraries	• UI doesn't feel 100% native • Performance limitations on older devices
	View Setup Guide	View Setup Guide	View Setup Guide

Mobile Dev ... 38% + Upgra Preview + Tools & files

Canvas Mobile Dev Hub .replit.dev /setup/react-native

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Dark Mode

React Native Setup Guide

Set up a modern React Native environment using Expo.

Installation Steps

1. Install Node.js

You need Node.js installed. We recommend using nvm (Node Version Manager).

```bash
nvm install 20
nvm use 20
```

2. Create the project

Use create-expo-app to scaffold a new project with TypeScript.

```bash
npx create-expo-app@latest my-app -t expo-template-blank-typescrip
cd my-app
```

3. Start the development server

This will start Metro bundler. You can open the app on a physical device using the Expo Go app, or press 'a' for Android emulator / 'i' for iOS simulator.

```bash
npx expo start
```

Prerequisites

- Node.js (v18+)
- Git
- Watchman (macOS only)

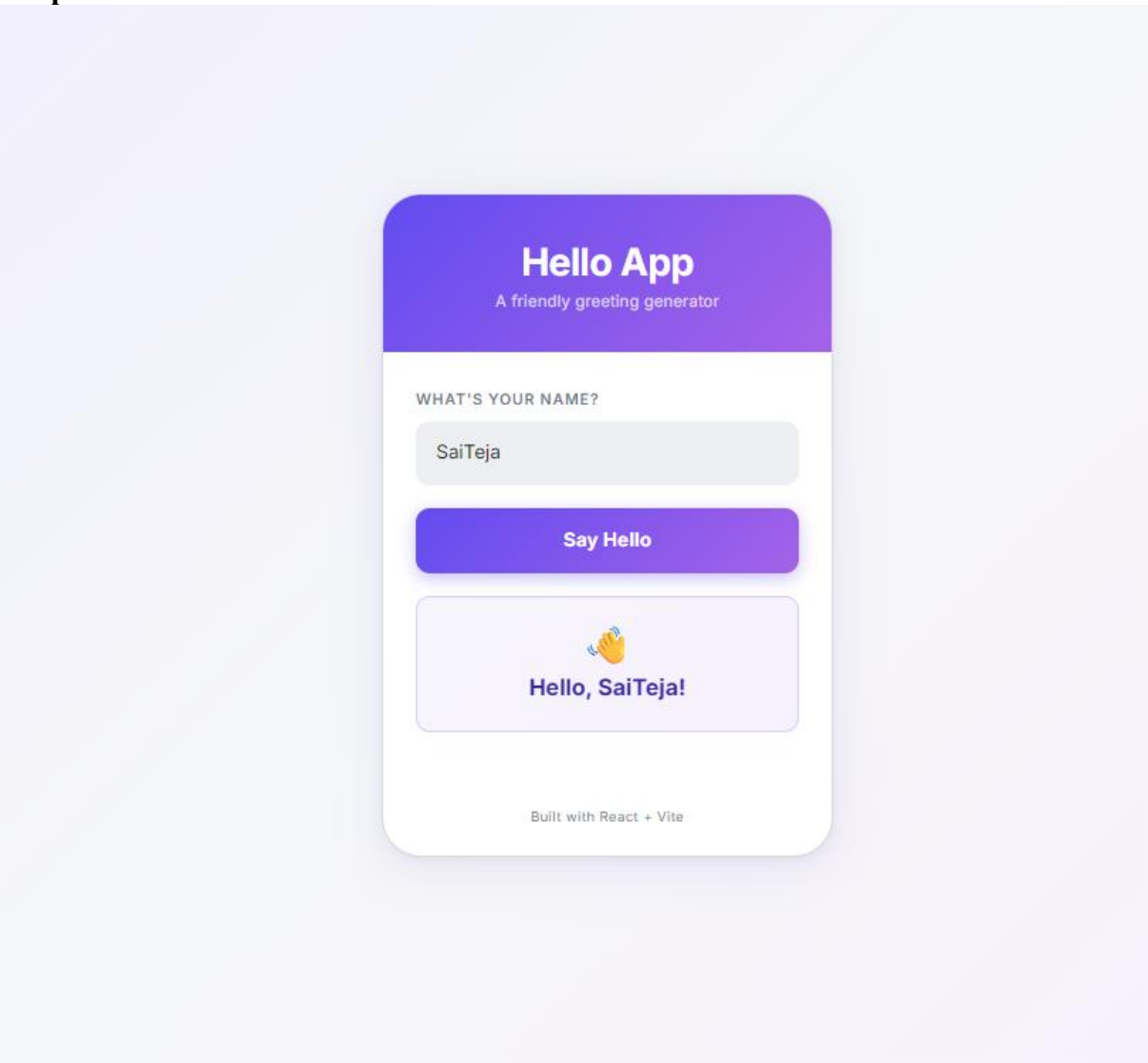
Gotchas

Make sure you have accepted all Android SDK licenses if you plan to build for Android. Run `yes | sdkmanager --licenses` in your terminal.

Task 2 – Basic Mobile App
• Build a simple "Hello App" that displays:
o App Title
o User's Name (input field + button)
o Greeting message dynamically updated.
• Use AI assistance to generate boilerplate code and explain structure.

Prompt:-
Create a basic mobile application called 'Hello App' using a suitable framework (React Native / Flutter / HTML5). The app should include an app title at the top, an input field to enter the user's name, and a button. When the button is clicked, a greeting message like 'Hello, [User Name]' should be displayed dynamically on the screen. Also, generate the boilerplate code for this application and explain the structure of the code in a simple and beginner-friendly way**.**

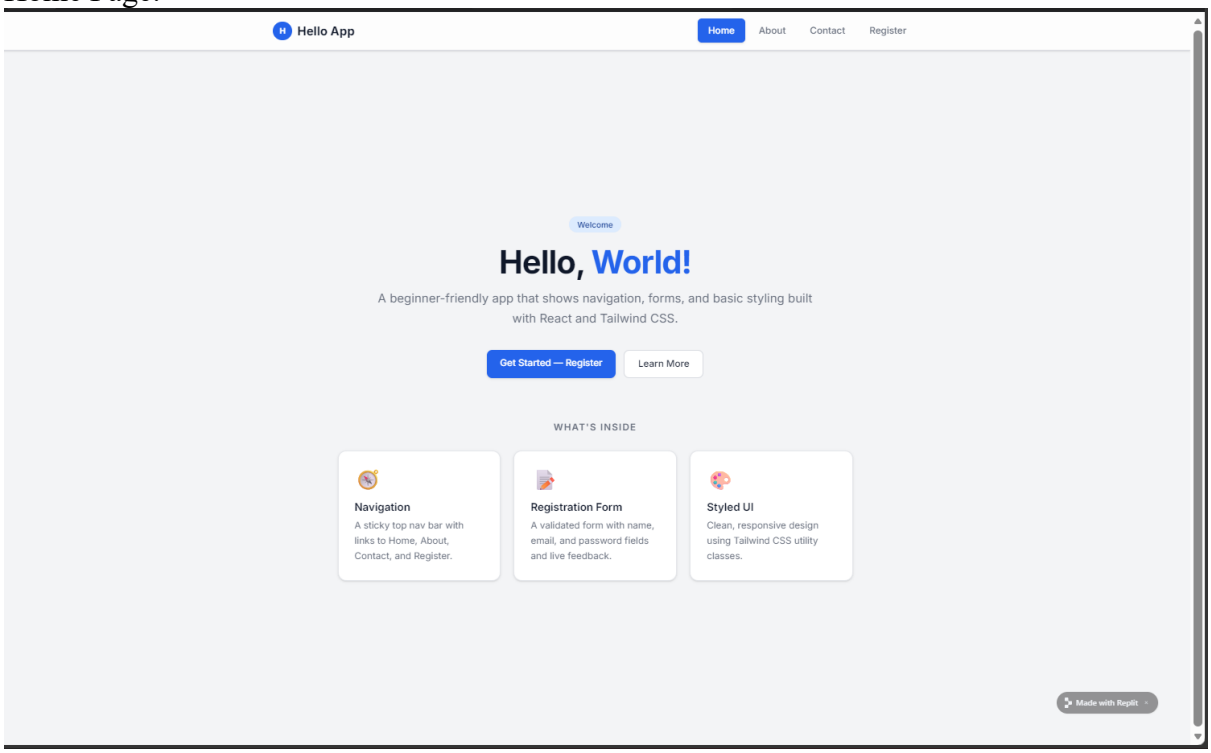
Output:-



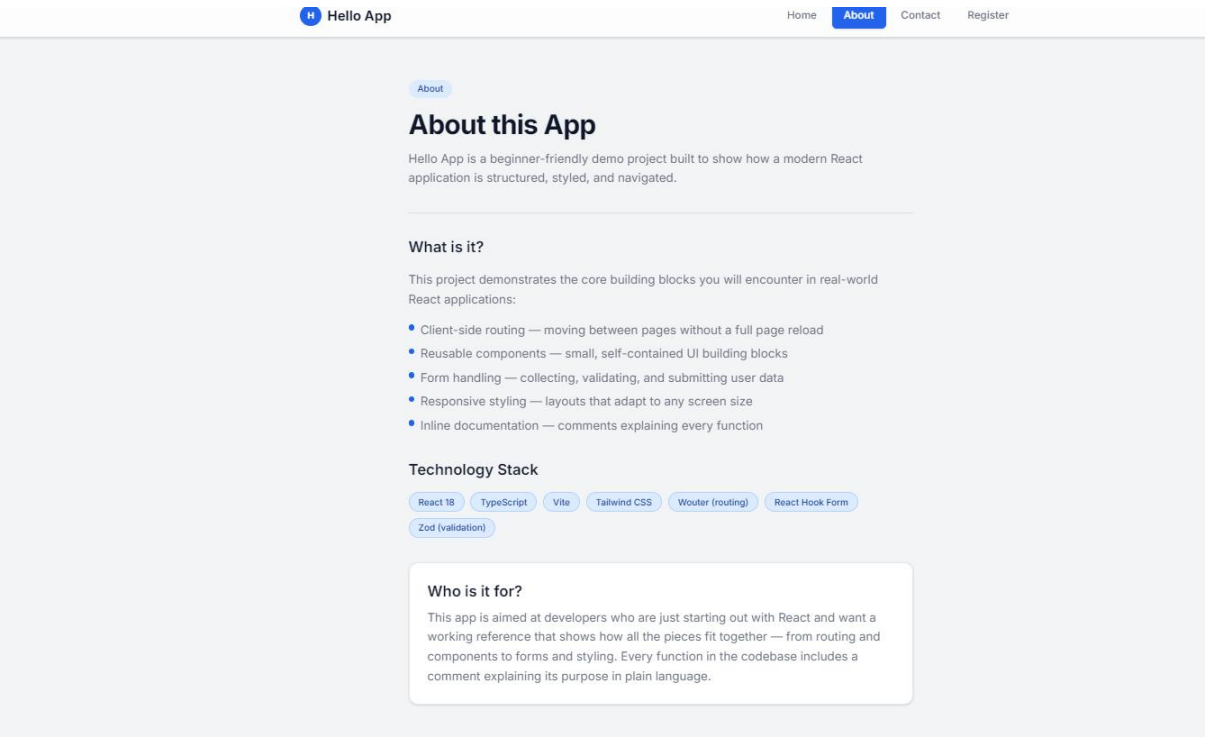
Task 3 – Adding Features with AI Assistance
• Enhance the app with:
o Navigation bar (Home, About, Contact).
o Form handling (e.g., user registration page).
o Basic styling with CSS or React Native styles.
• Ask AI in Replit to add inline documentation for each function.

Prompt:-
enhanced the app by adding a navigation bar with pages like Home, About, and Contact. I also created a basic user registration form with simple validation. To improve the appearance, I applied basic styling to the app. Additionally, I included inline comments in the code to make it easier to understand each function.

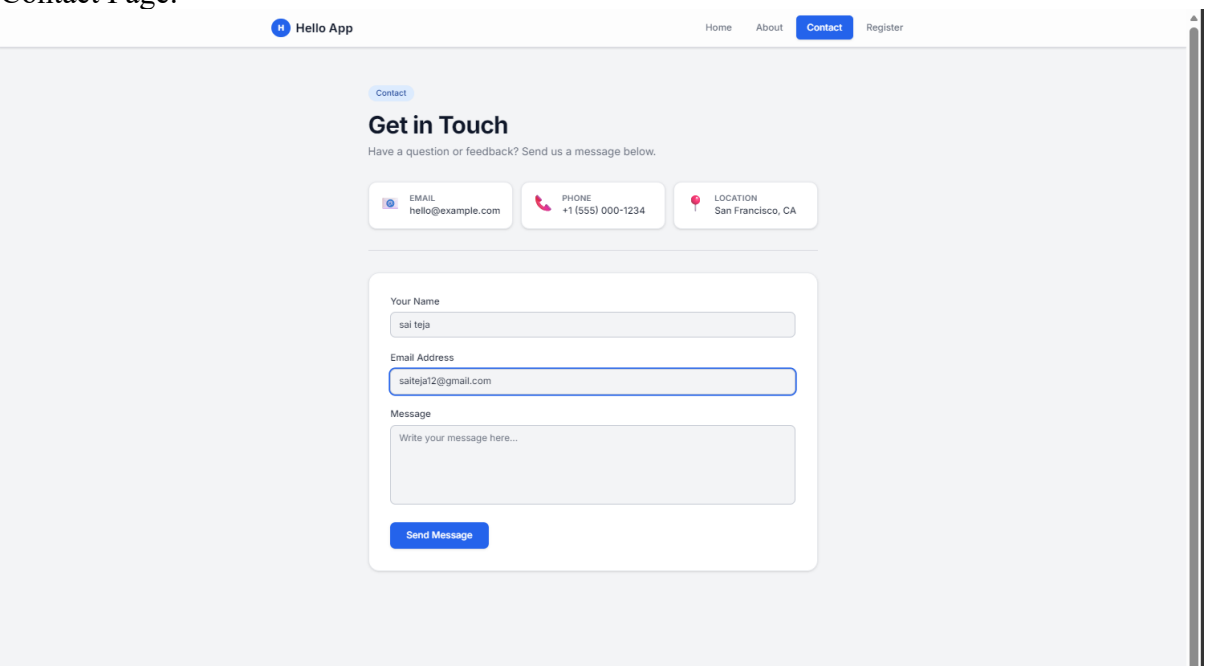
Output:-
Home Page:



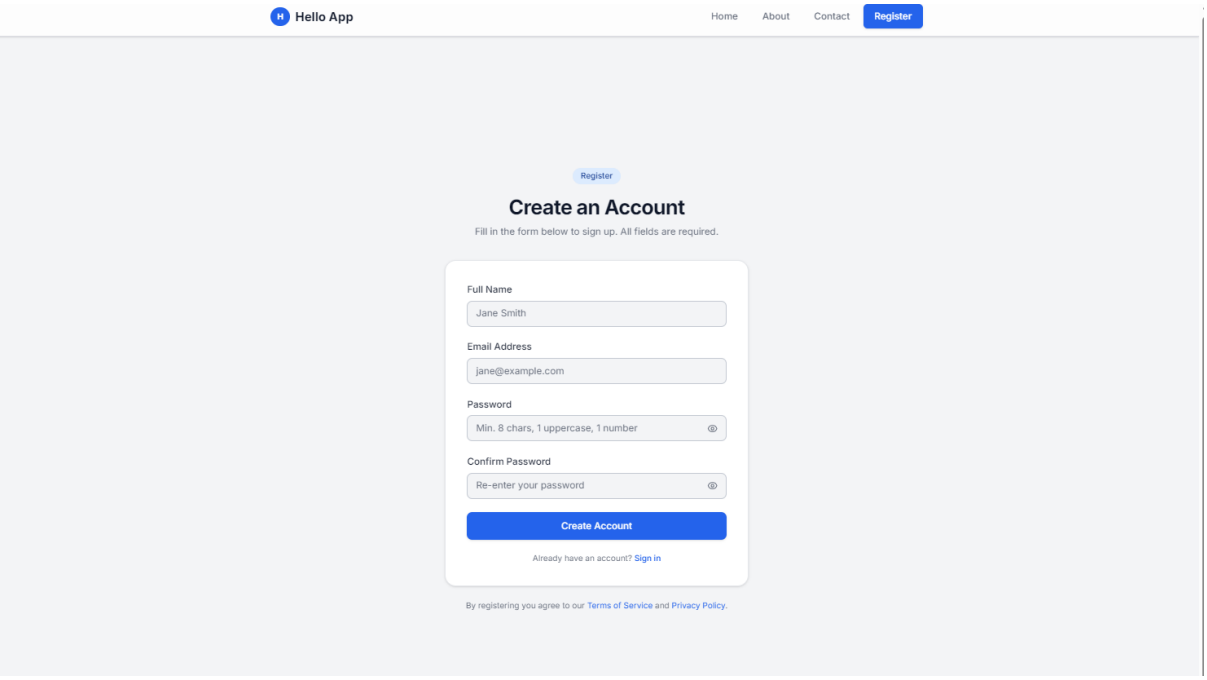
About page:



Contact Page:



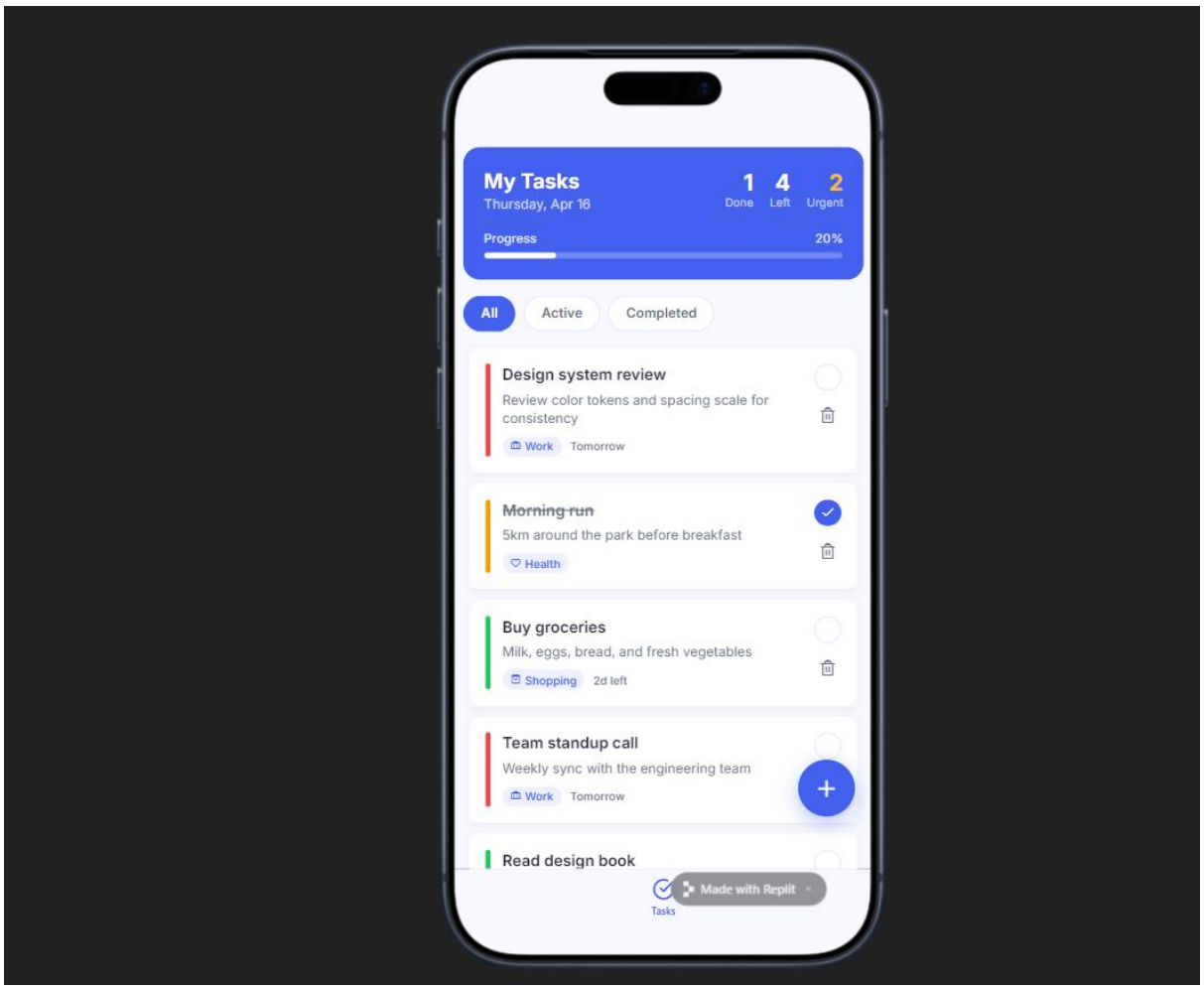
Registration Page:



Task 4 – Testing on Mobile
• Run the app in Replit's mobile preview mode.
• Test responsiveness on different screen sizes.
• Fix UI alignment issues with AI suggestions.

Prompt:-
Enhance the existing 'Hello App' by adding more features using a suitable framework (React Native / Flutter / HTML5). Include a navigation bar with options like Home, About, and Contact pages. Implement form handling by creating a user registration page with input fields (such as name, email, and password) and a submit button. Apply basic styling using CSS or React Native styles to improve the app's appearance. Also, generate the updated code and add inline documentation/comments for each function to explain its purpose in a simple and beginner-friendly way.

Output:-
Home Page:

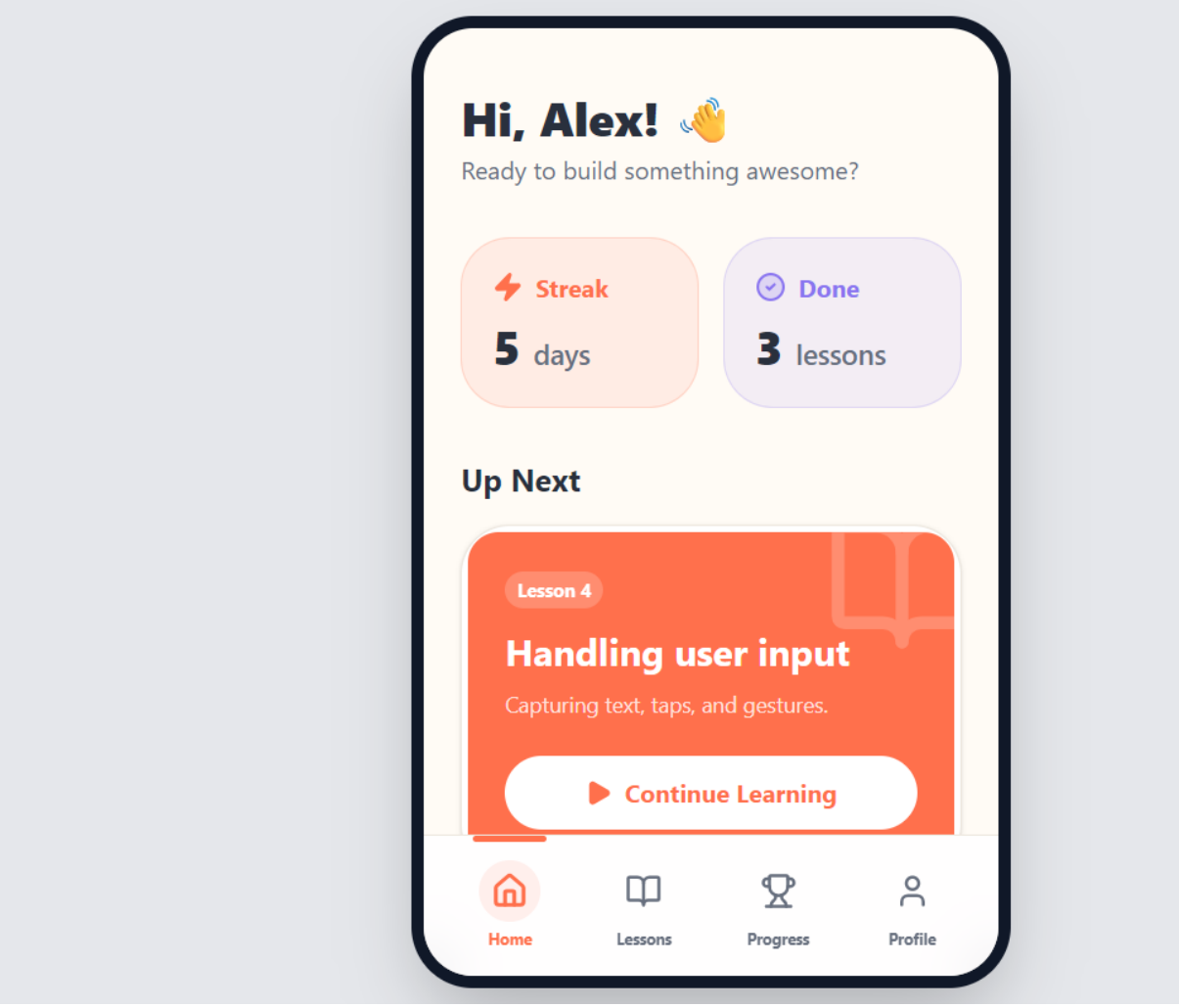


Task 5 – Deployment
• Deploy the app using Replit hosting or export it to a mobile framework (like Expo for React Native).
• Share the deployment link with peers for testing.
Expected Outcomes:
• Students will understand how to use Replit for mobile app development.
• Ability to build a basic functional mobile app with UI, navigation, and user interaction.
• Experience in AI-assisted coding, debugging, and documentation for mobile applications.

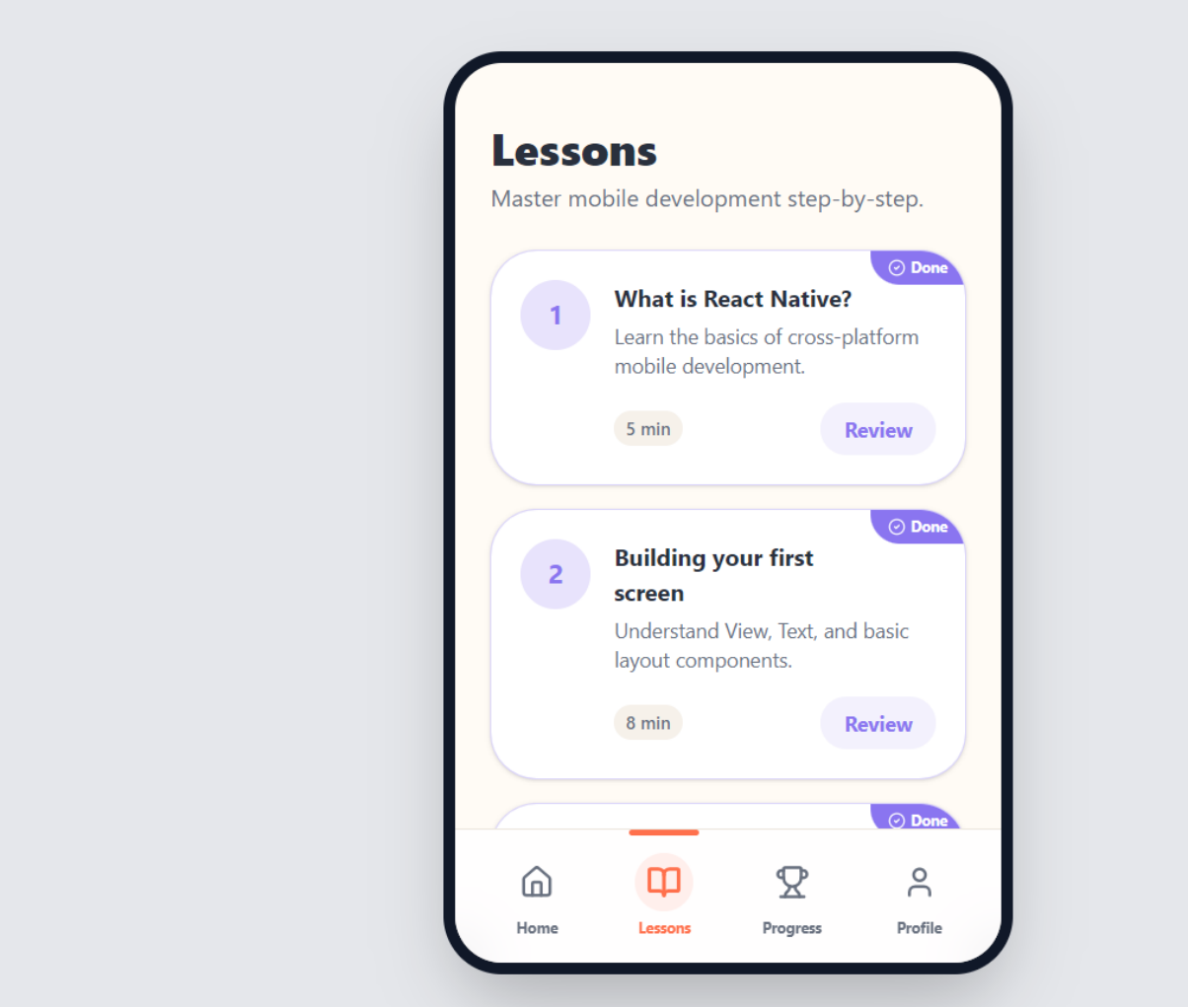
Prompt:-
Deploy the completed mobile application using Replit hosting or export it to a mobile development framework such as Expo (for React Native). Provide step-by-step instructions for deployment and generate a working deployment link that can be shared with others for testing. Also, explain the expected outcomes, including understanding how to use Replit for mobile app development, the ability to build a functional mobile app with UI, navigation, and user interaction, and gaining experience in AI-assisted coding, debugging, and documentation in a simple and beginner-friendly way.

Ouptut:-

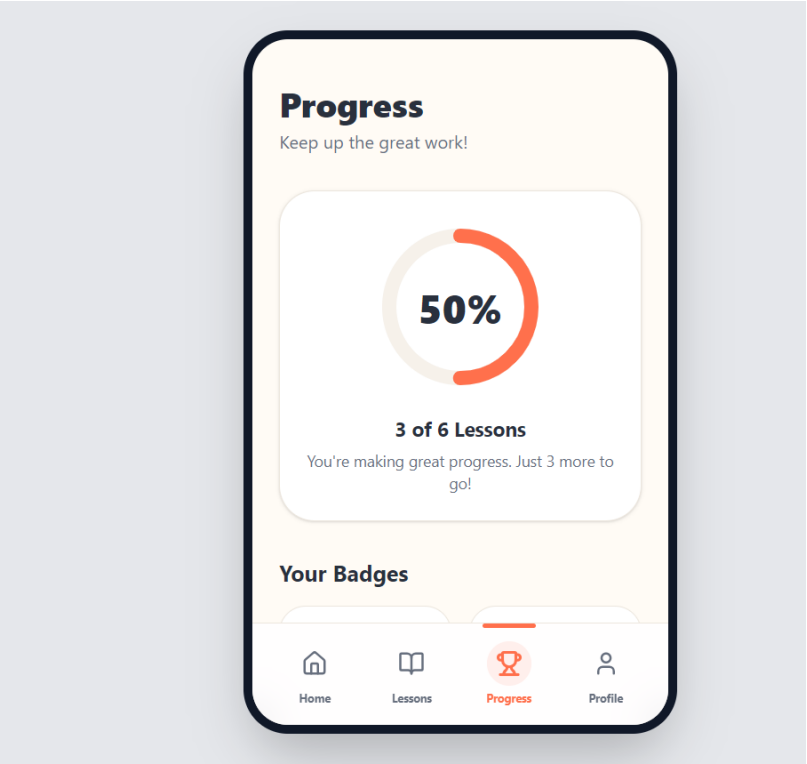
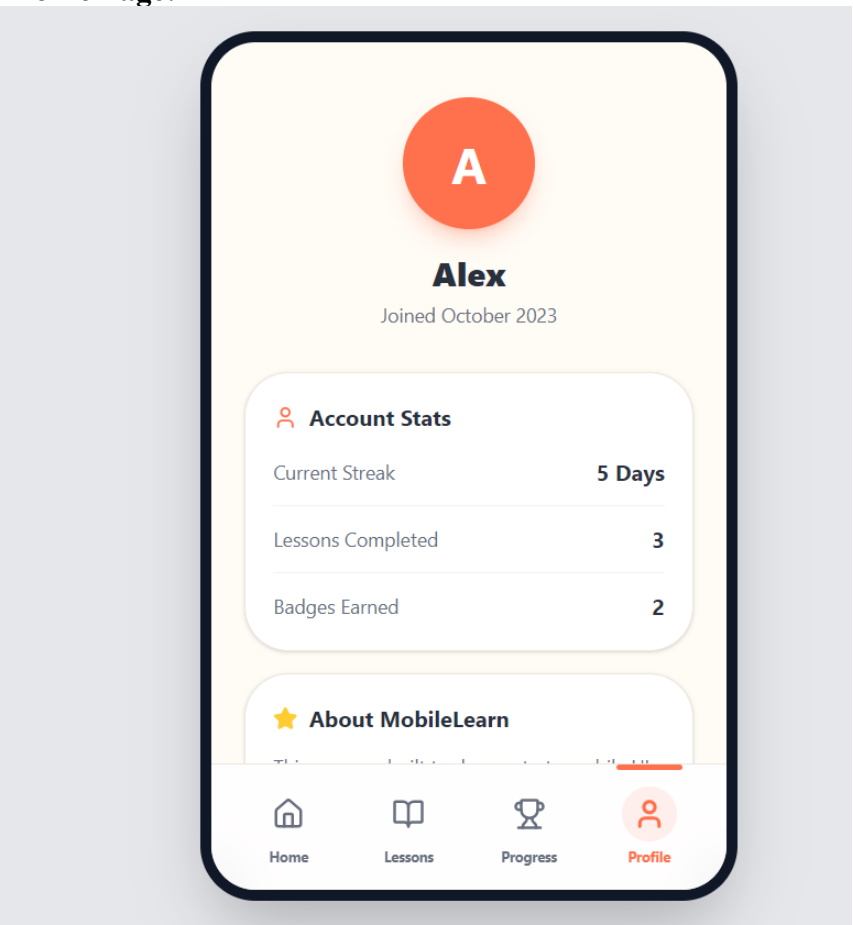
Home page:



Lessons page:



Progress Page:-



Profile Page:-